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Registration no : 23-BCS-065

Course : DevOps

Department : CS

Semester : 7th(2023-2027)

Submitted to : Mr Waqas Saleem

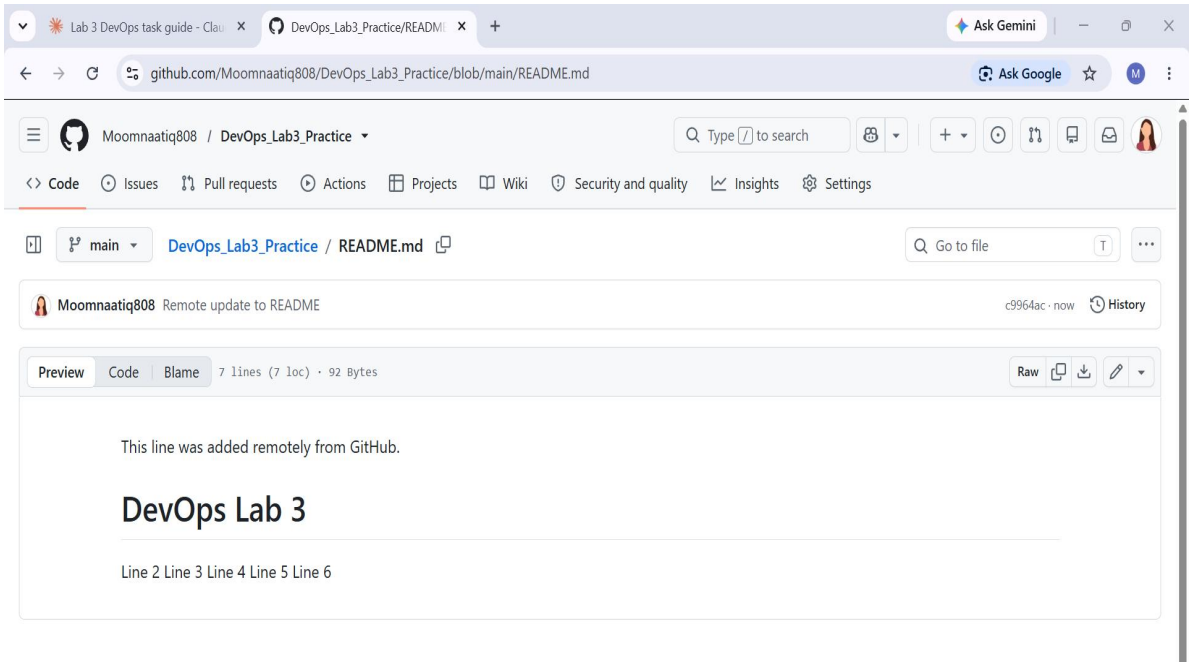
Lab#03:

Working with Git History, Stashing, and Reverting Commits

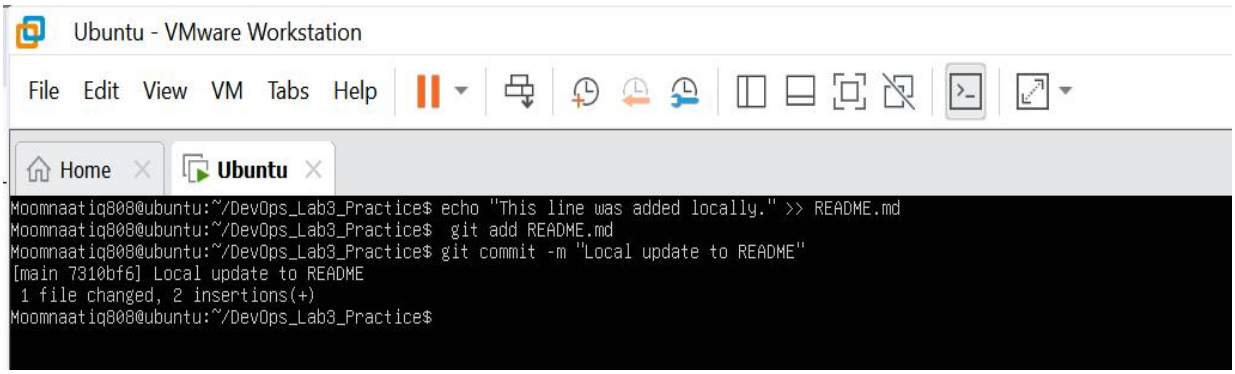
Lab 3 – Working with Git History, Stashing, and Reverting Commits

Task 1 – Handling Local and Remote Commit Conflicts (Pull vs Pull --rebase

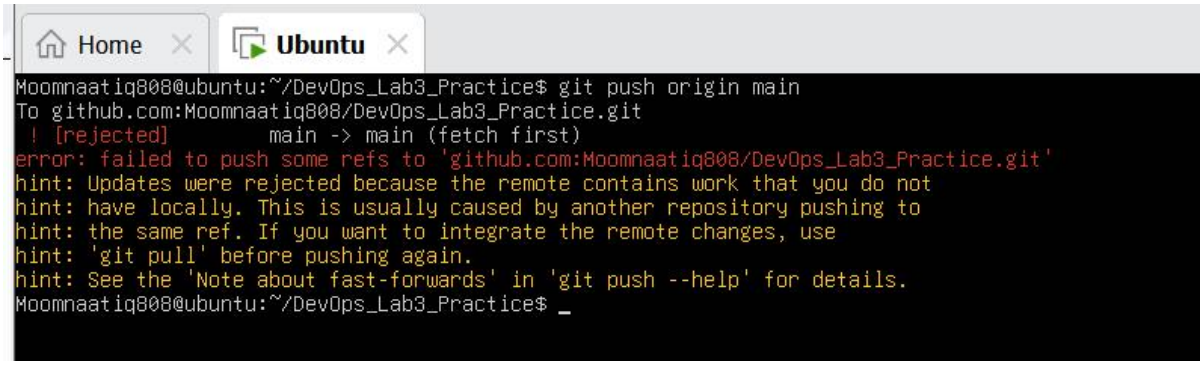
remote_edit.png



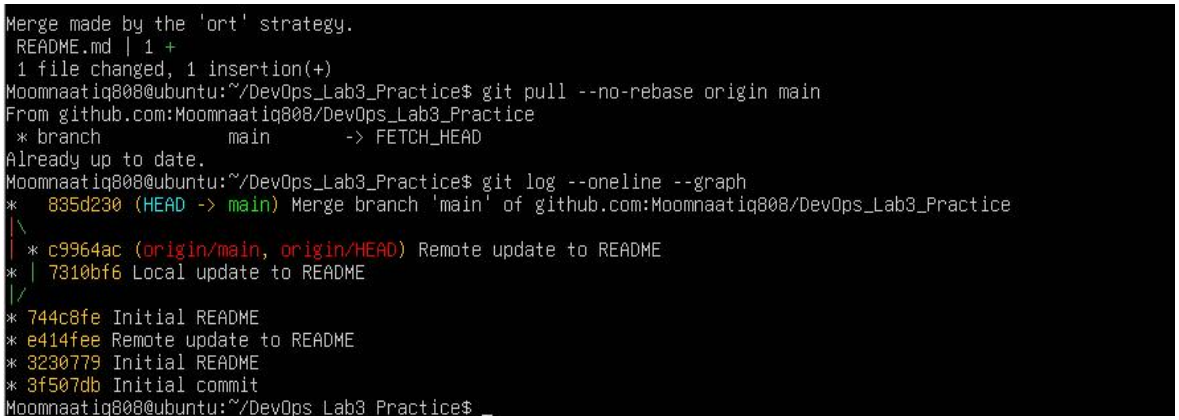
local_commit.png



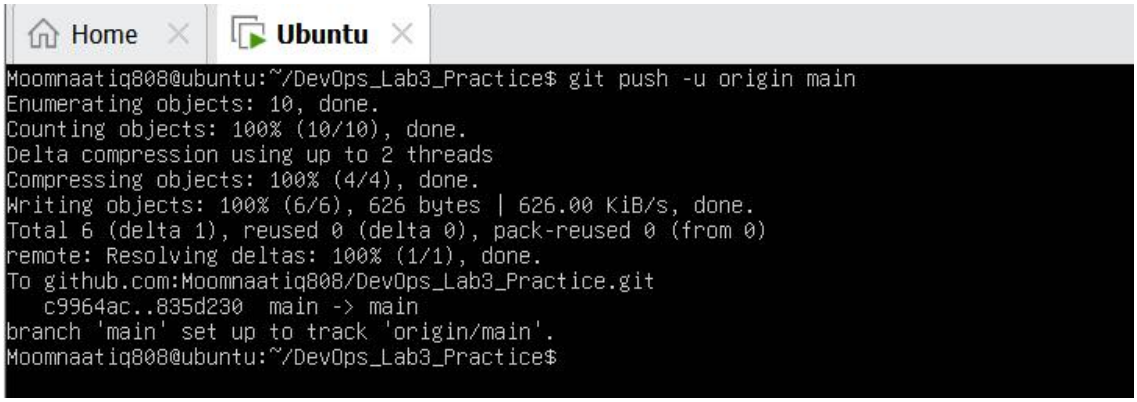
push_error.png



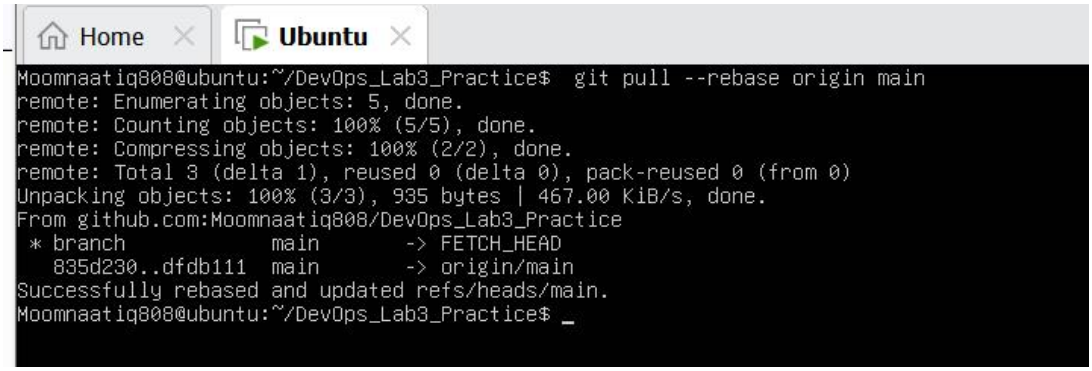
merge_commit.png



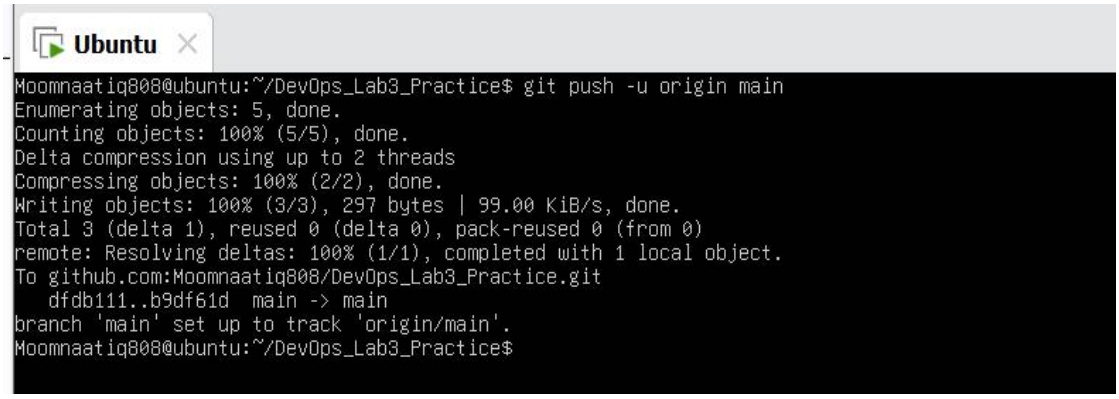
push_after_merge.png



rebase_pull.png

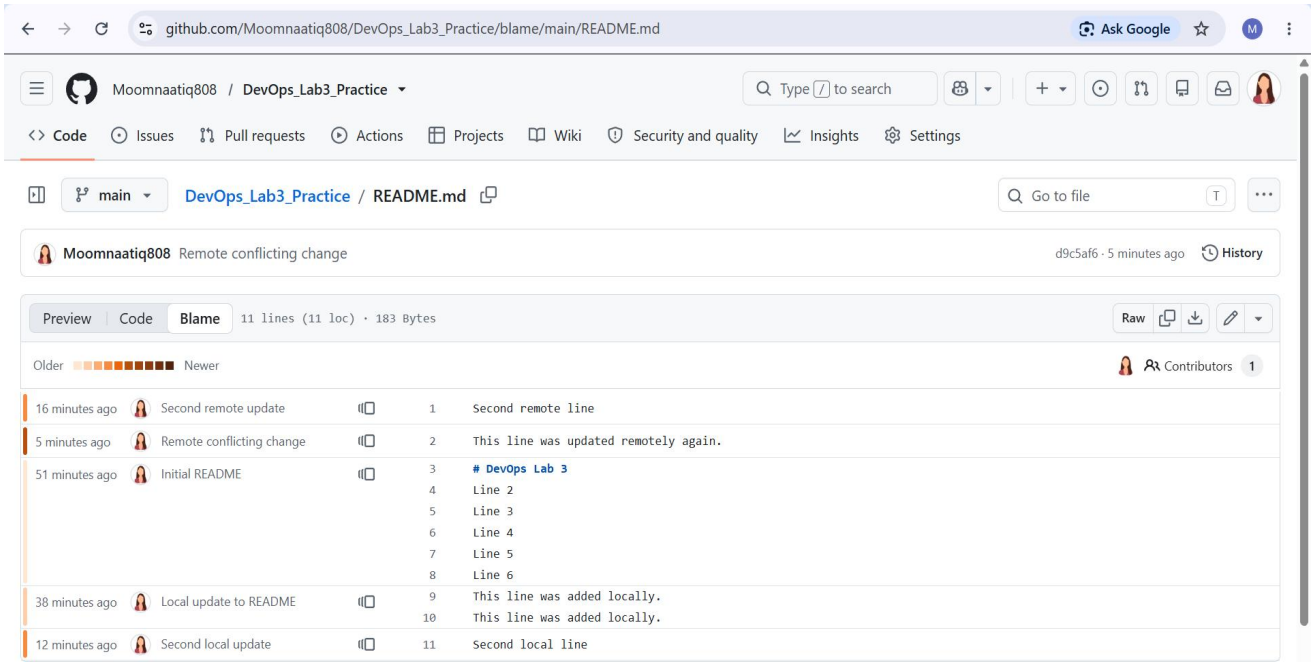


push_after_rebase.png

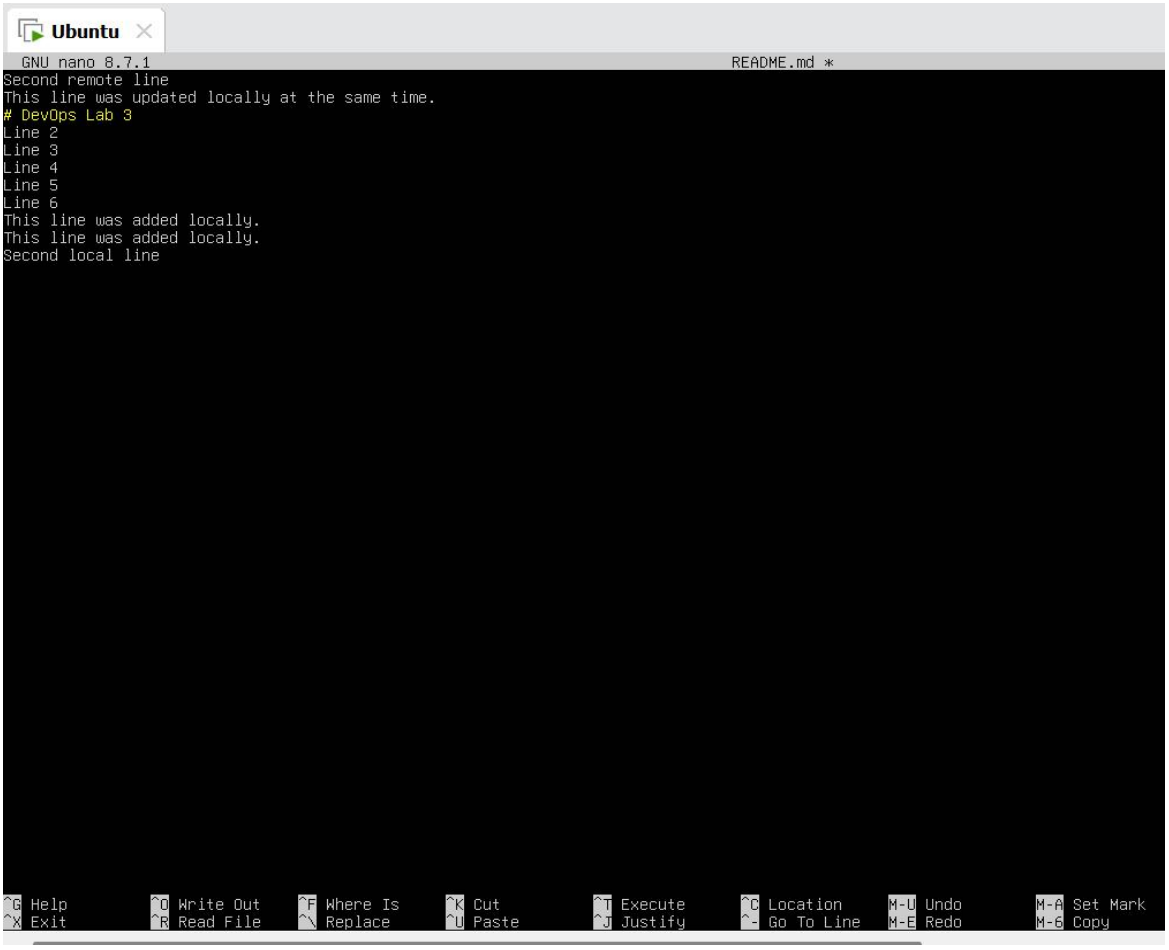


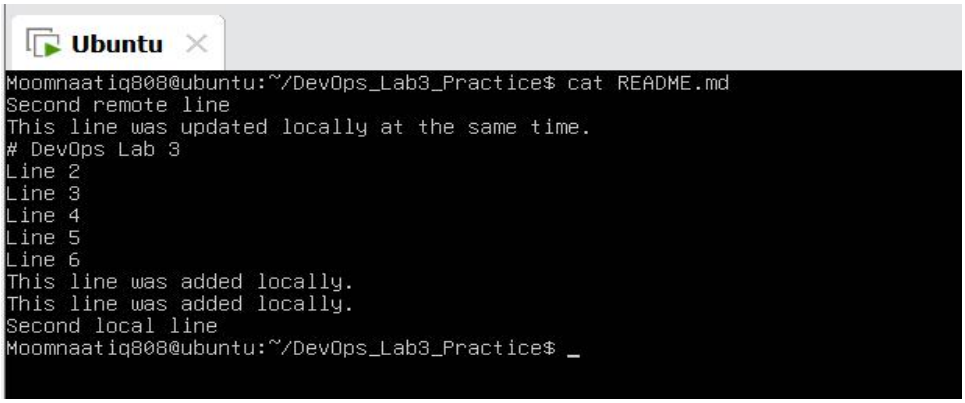
Task 2 – Creating and Resolving Merge Conflicts Manually

remote_conflict_edit.png

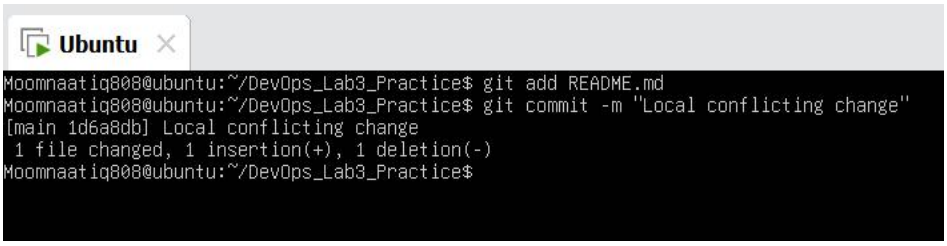


local_conflict_edit.png

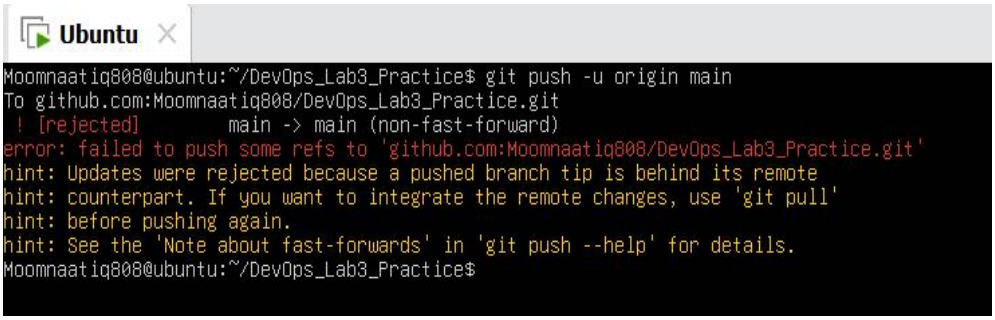




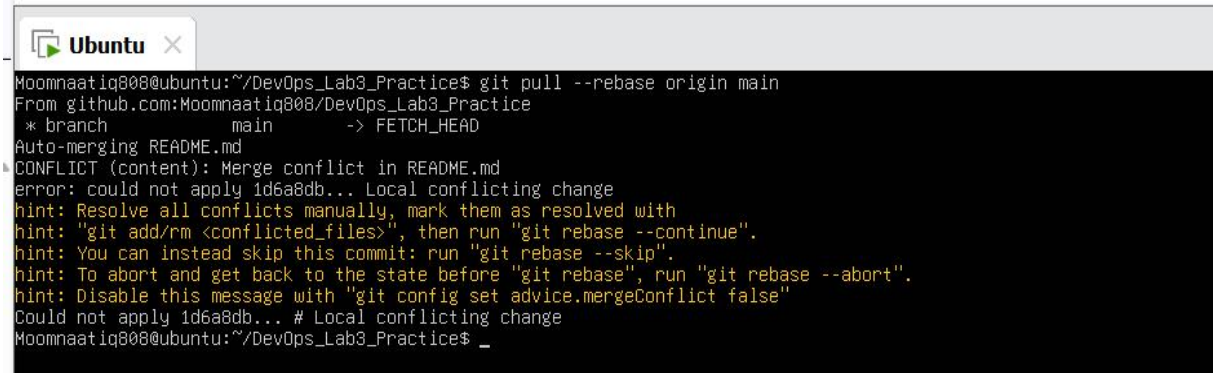
local_conflict_commit.png



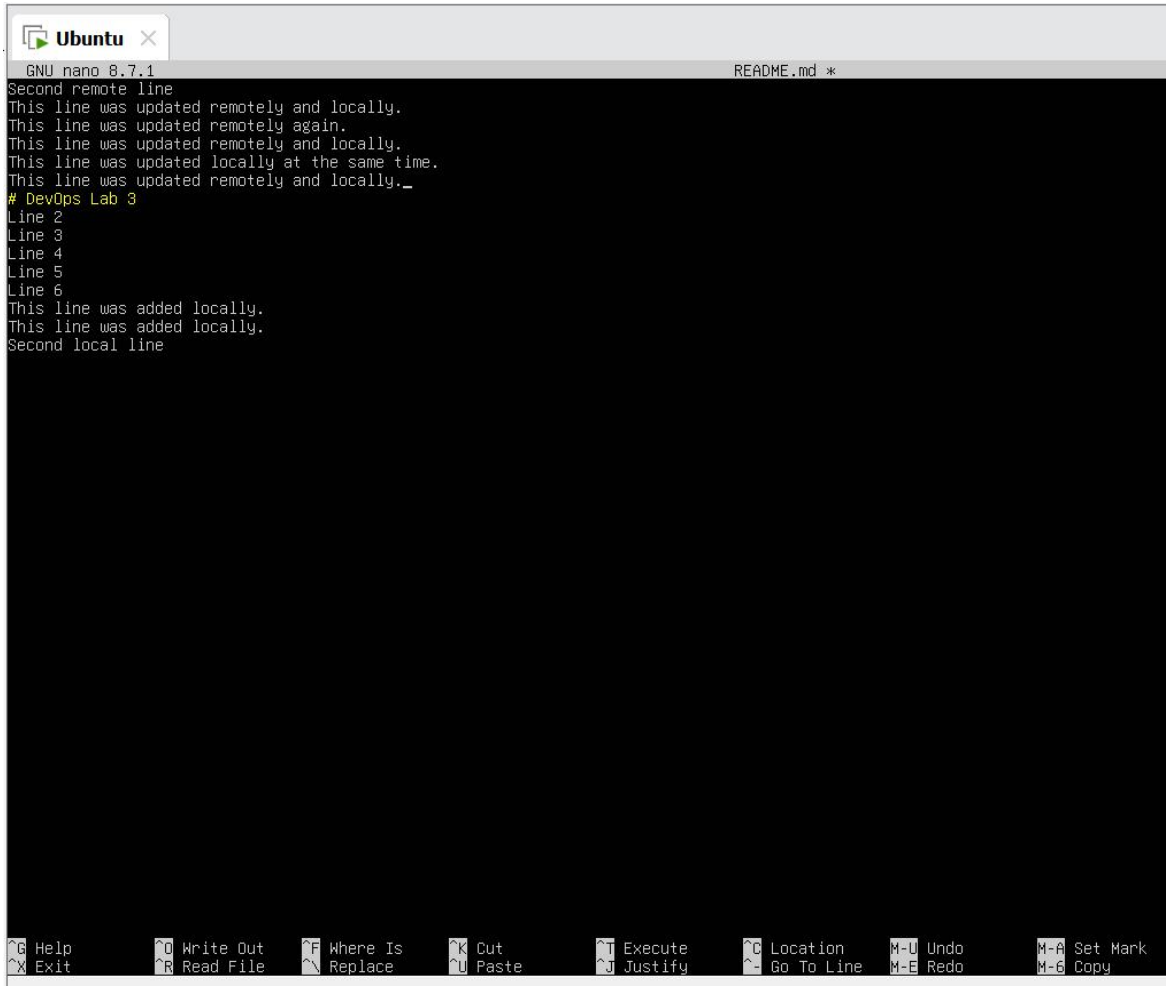
conflict_push_error.png



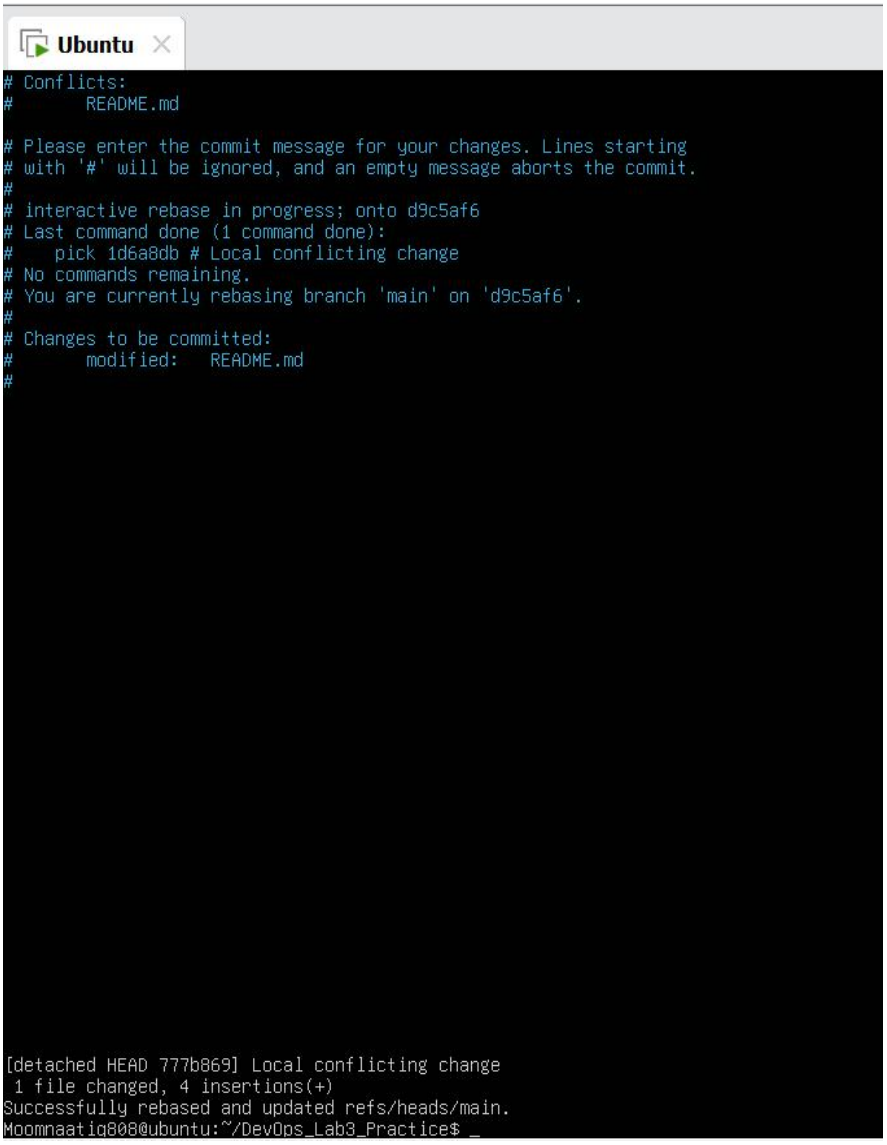
conflict_message.png



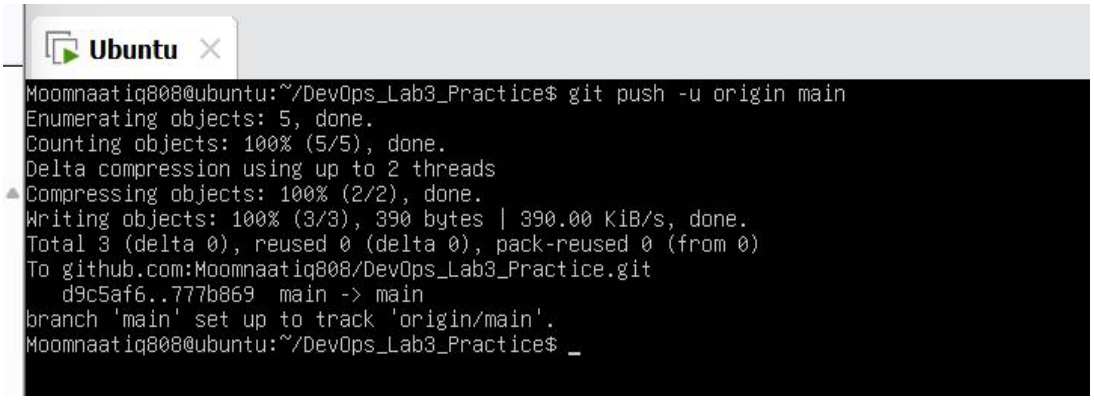
resolved_readme.png



rebase_continue.png

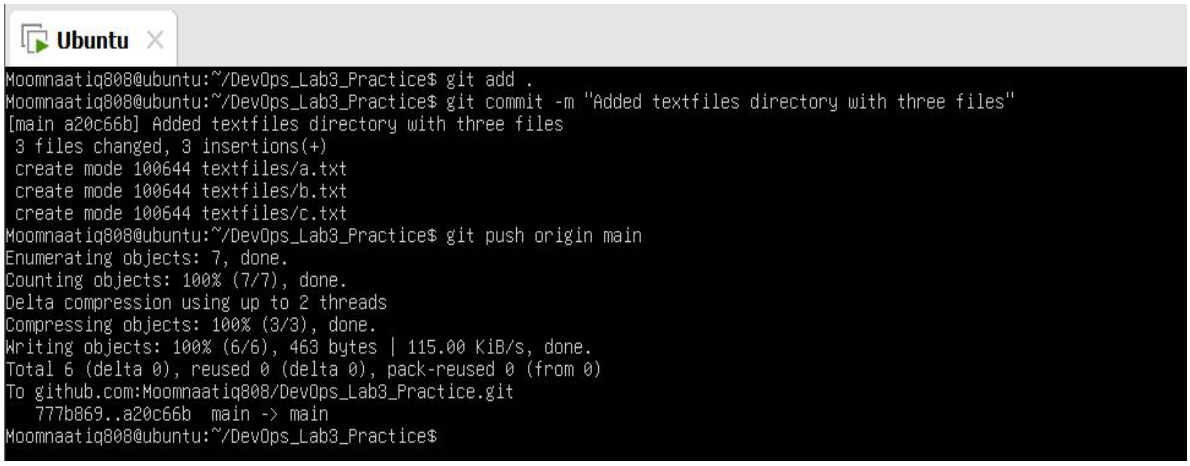


push_after_resolve.png

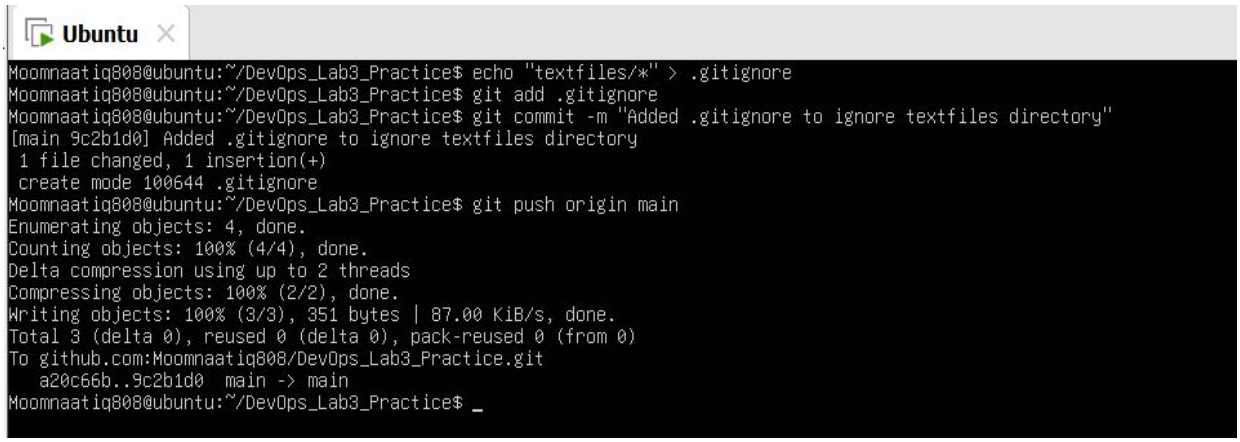


Task 3 – Managing Ignored Files with .gitignore and Removing Tracked Files

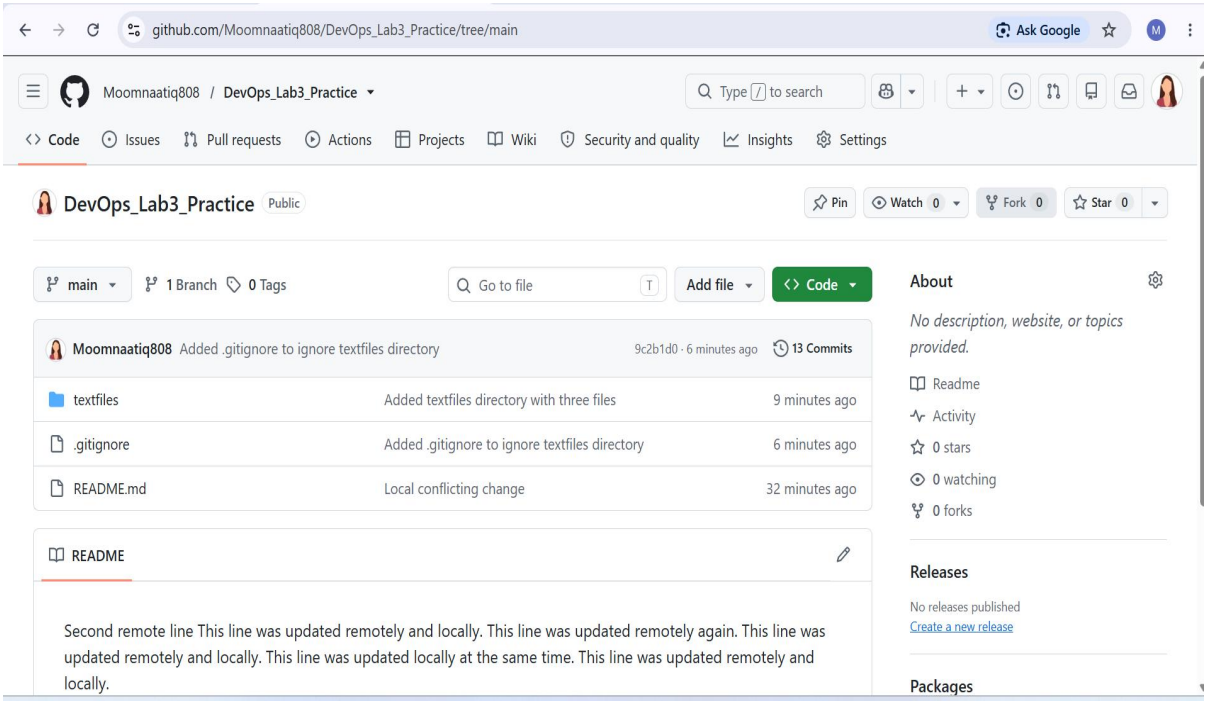
push_textfiles.png



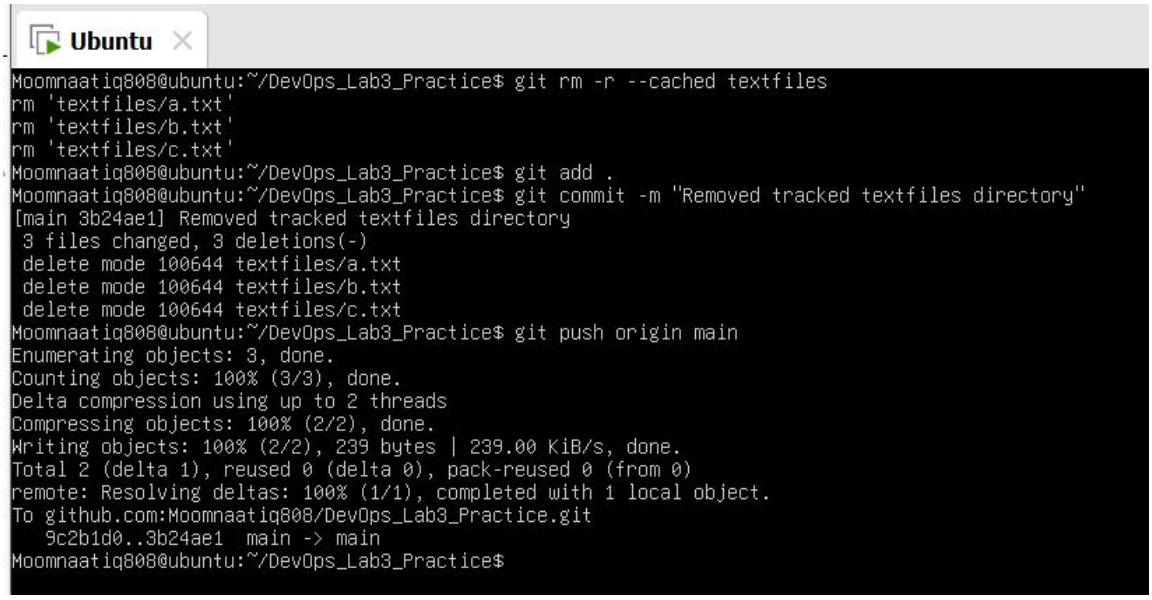
gitignore_push.png



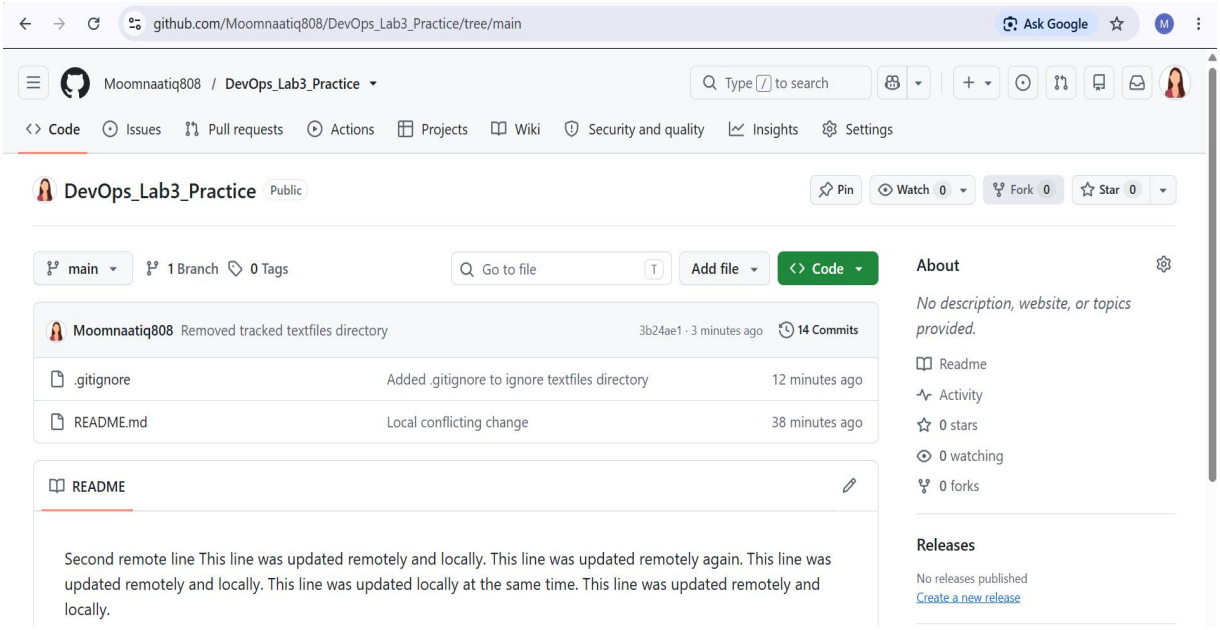
repo_still_has_textfiles.png



rm_cached_push.png

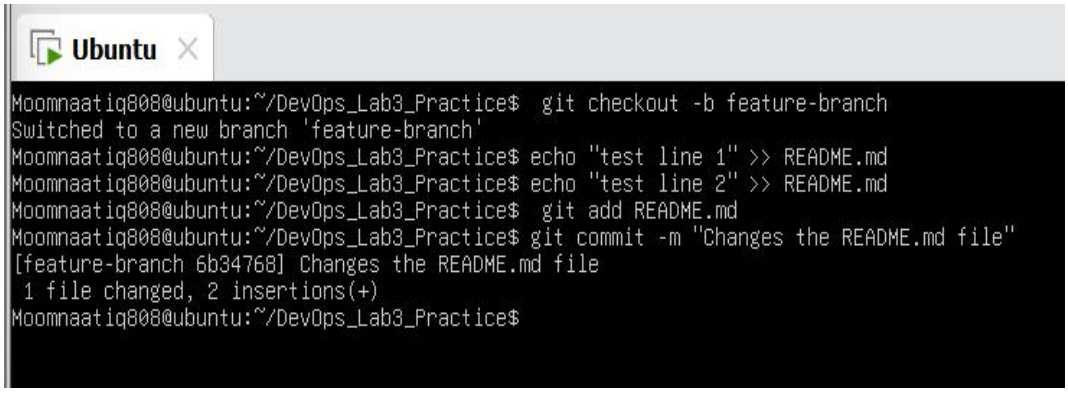


repo_textfiles_removed.png

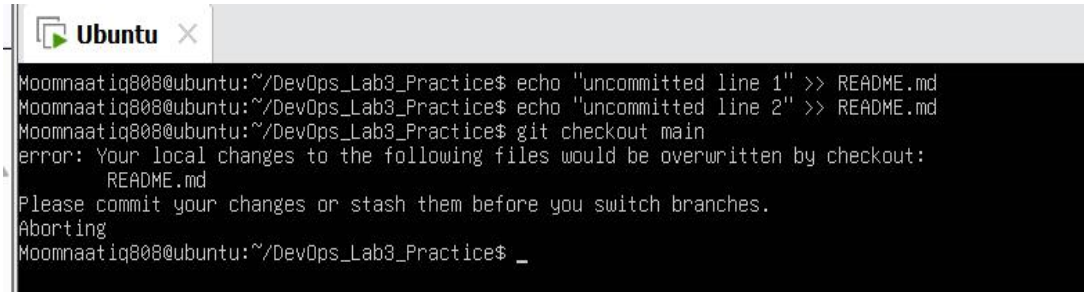


Task 4 – Create Temporary Changes and Use git stash

modified_readme.png



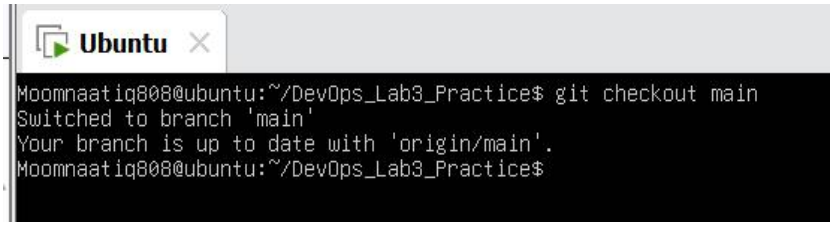
checkout_error.png



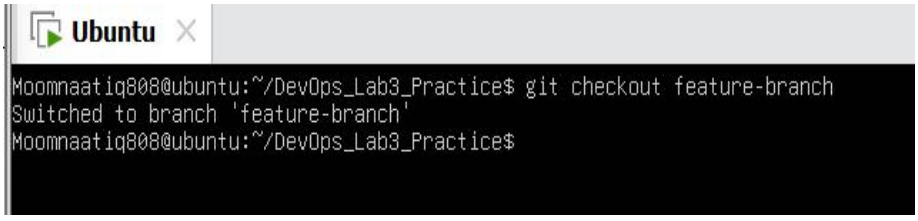
stash_command.png



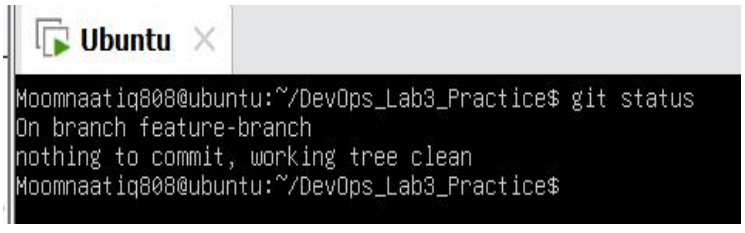
branch_switched.png



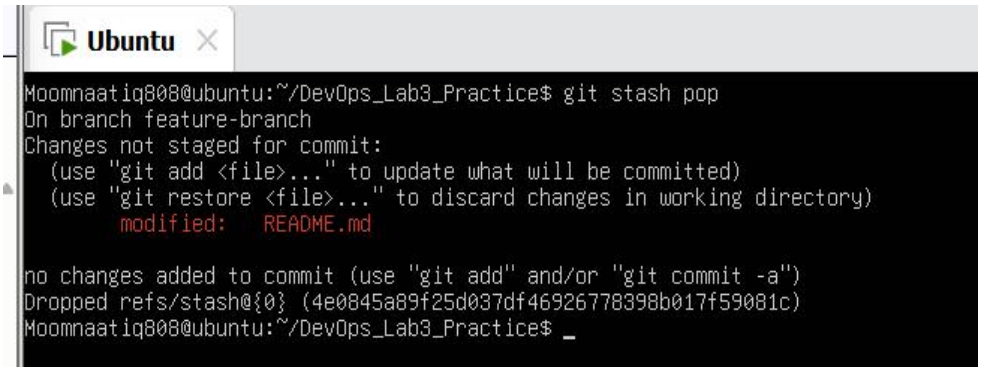
back_to_feature.png



status_clean.png

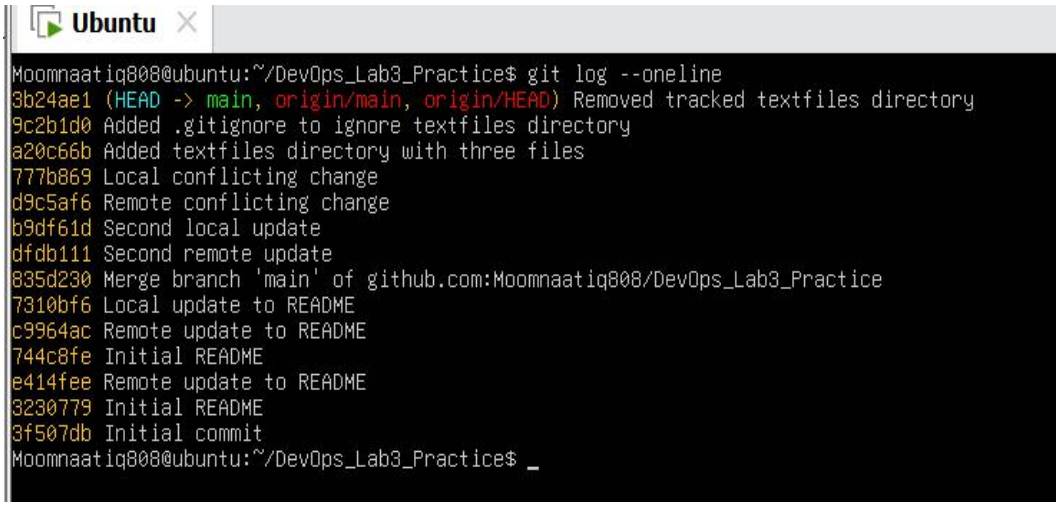


stash_pop.png

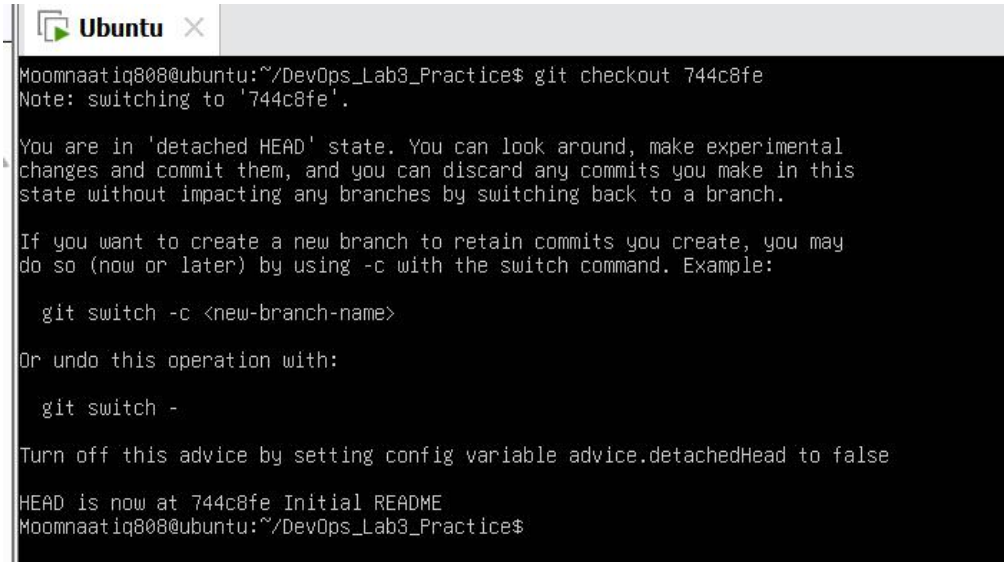


Task 5 – Checkout a Specific Commit Using git log

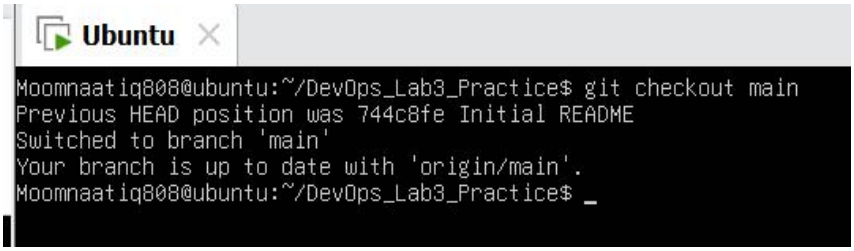
log_before_checkout.png



detached_head.png

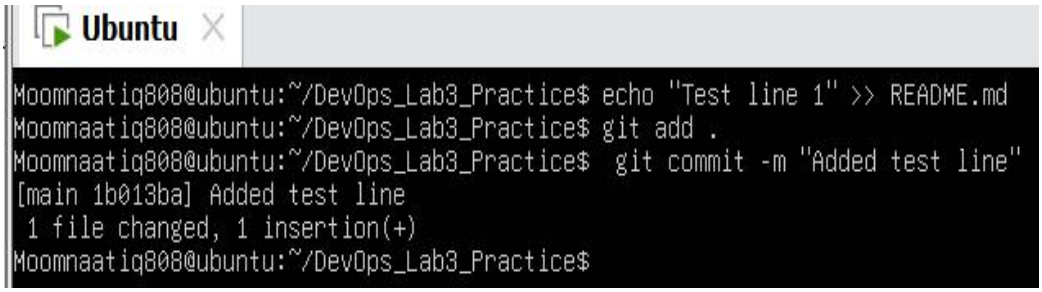


back_to_main.png

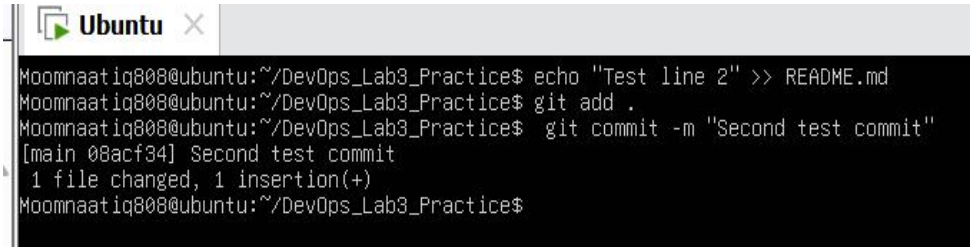


Task 6 – Resetting Commits (Soft vs Hard Reset) (With Verification Steps)

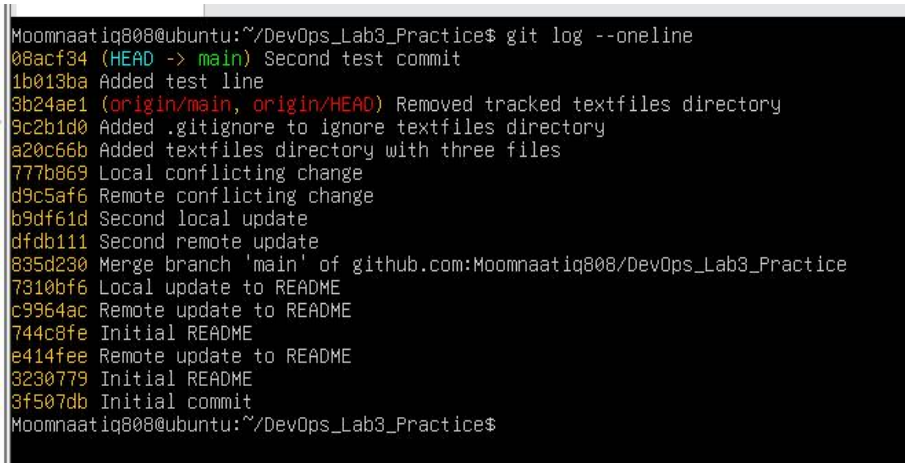
first_commit.png



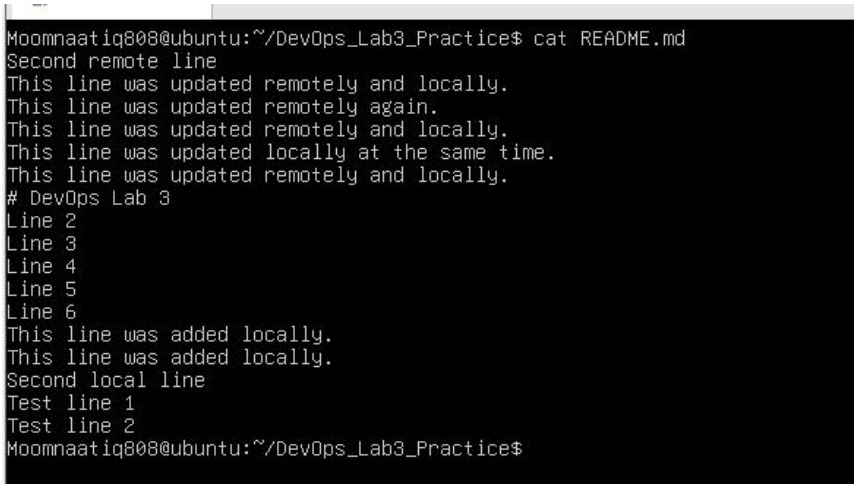
second_commit.png



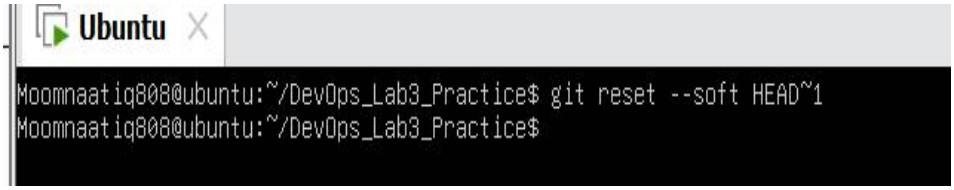
log_before_reset.png



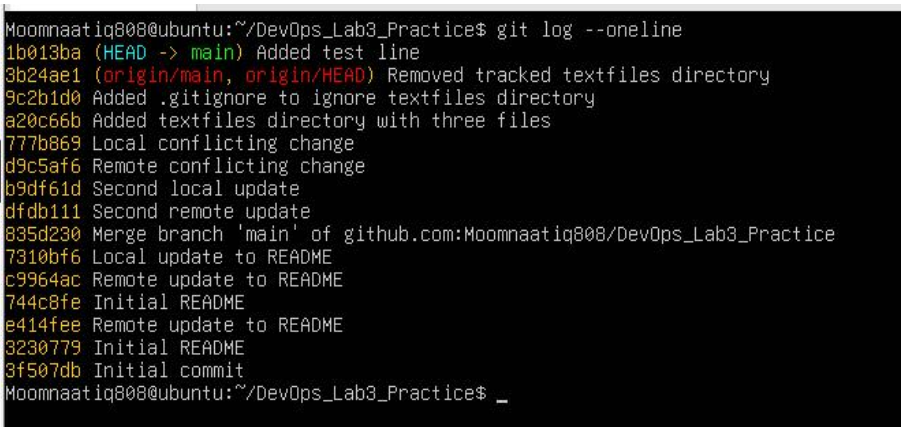
file_before_reset.png



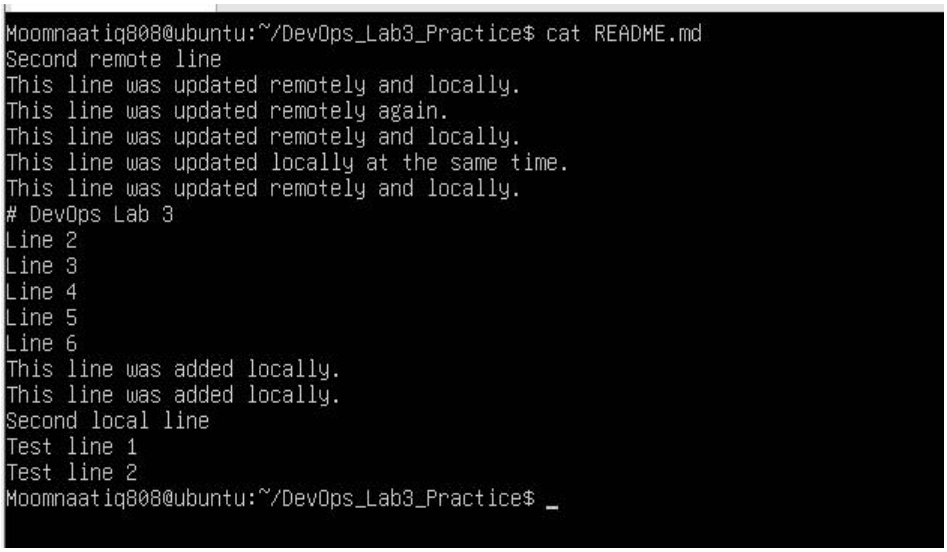
soft_reset.png



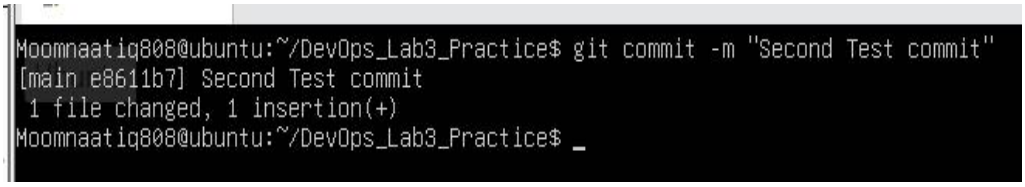
log_after_soft_reset.png



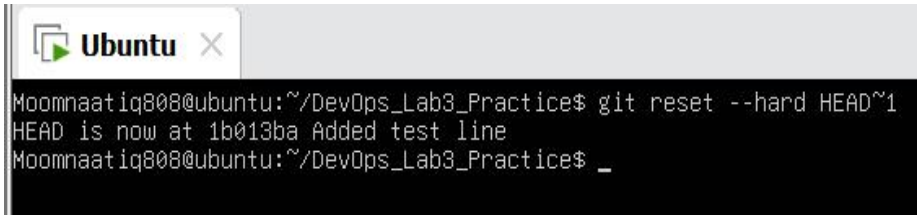
file_after_soft_reset.png



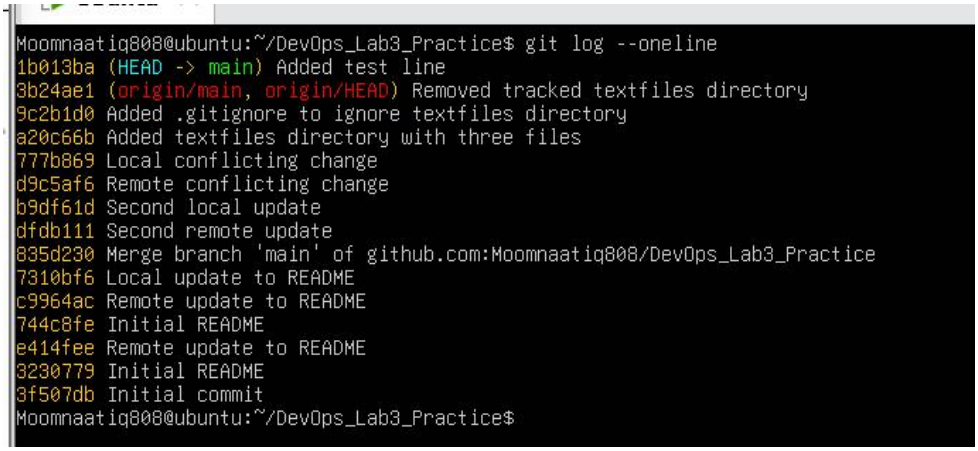
recommit_after_soft_reset.png



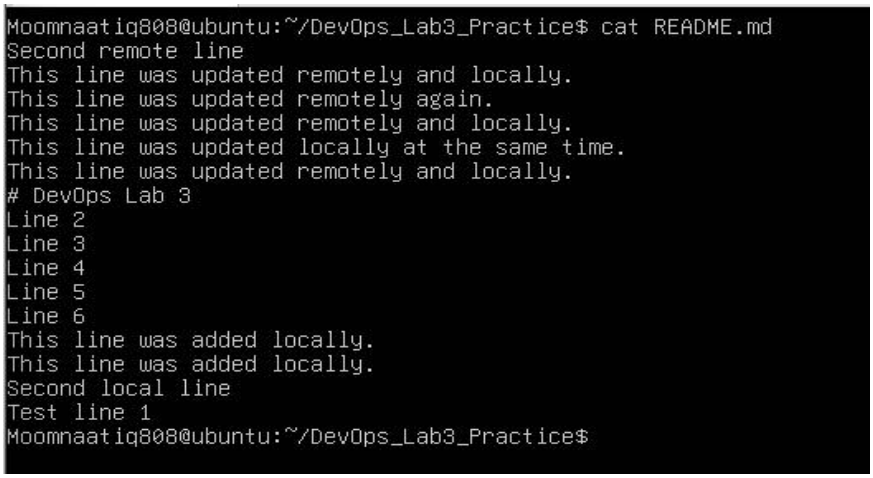
hard_reset.png



log_after_hard_reset.png



file_after_hard_reset.png

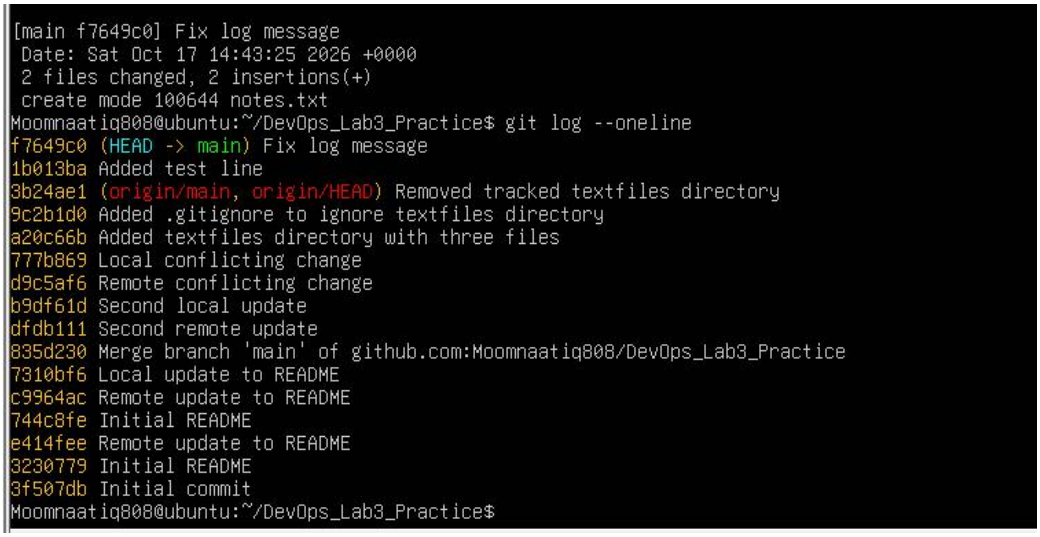


Task 7 – Amending the Last Commit

first_amend_commit.png



amend_commit.png



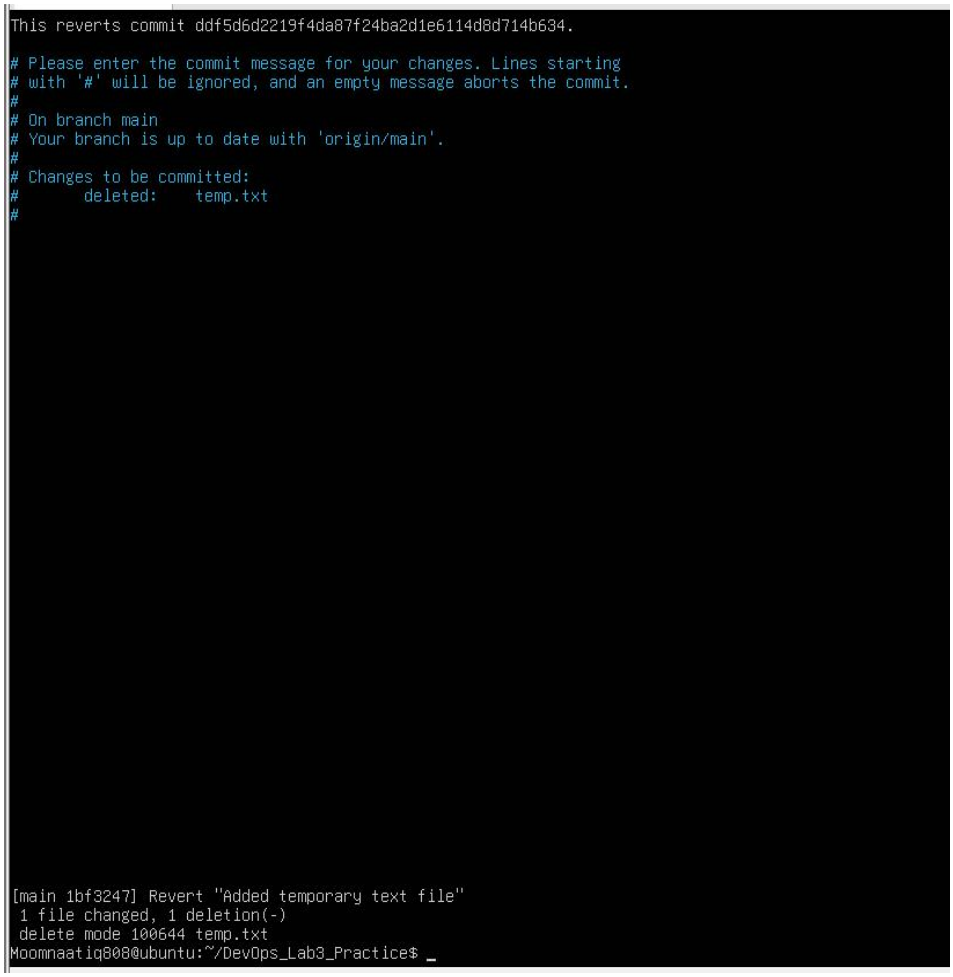
Task 8 – Reverting a Commit (Safe Undo on Remote Branch)

commit_temp_file.png

```
Moomnaatiq808@ubuntu:~/DevOps_Lab3_Practice$ echo "temporary text" >> temp.txt
Moomnaatiq808@ubuntu:~/DevOps_Lab3_Practice$ git add .
Moomnaatiq808@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Added temporary text file"
[main ddf5d6d] Added temporary text file
 1 file changed, 1 insertion(+)
 create mode 100644 temp.txt
Moomnaatiq808@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 2 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (10/10), 896 bytes | 34.00 KiB/s, done.
Total 10 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To github.com:Moomnaatiq808/DevOps_Lab3_Practice.git
   3b24ae1..ddf5d6d  main -> main
Moomnaatiq808@ubuntu:~/DevOps_Lab3_Practice$
```

revert_commit.png

```
Moomnaatiq808@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
1bf3247 (HEAD -> main) Revert "Added temporary text file"
ddf5d6d (origin/main, origin/HEAD) Added temporary text file
f7649c0 Fix log message
1b013ba Added test line
3b24ae1 Removed tracked textfiles directory
9c2b1d0 Added .gitignore to ignore textfiles directory
a20c66b Added textfiles directory with three files
777b869 Local conflicting change
d9c5af6 Remote conflicting change
b9df61d Second local update
dfdb111 Second remote update
835d230 Merge branch 'main' of github.com:Moomnaatiq808/DevOps_Lab3_Practice
7310bf6 Local update to README
c9964ac Remote update to README
744c8fe Initial README
e414fee Remote update to README
3230779 Initial README
3f507db Initial commit
Moomnaatiq808@ubuntu:~/DevOps_Lab3_Practice$
```



revert_push.png

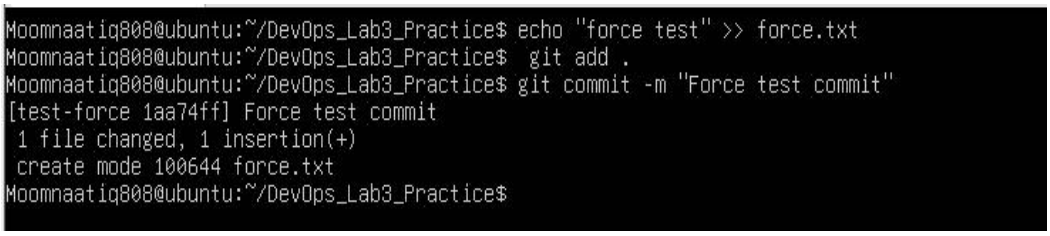


Task 9 – Force Push (With Caution)

new_branch.png



force_commit.png



push_force_branch.png



hard_reset_force.png



normal_push.png

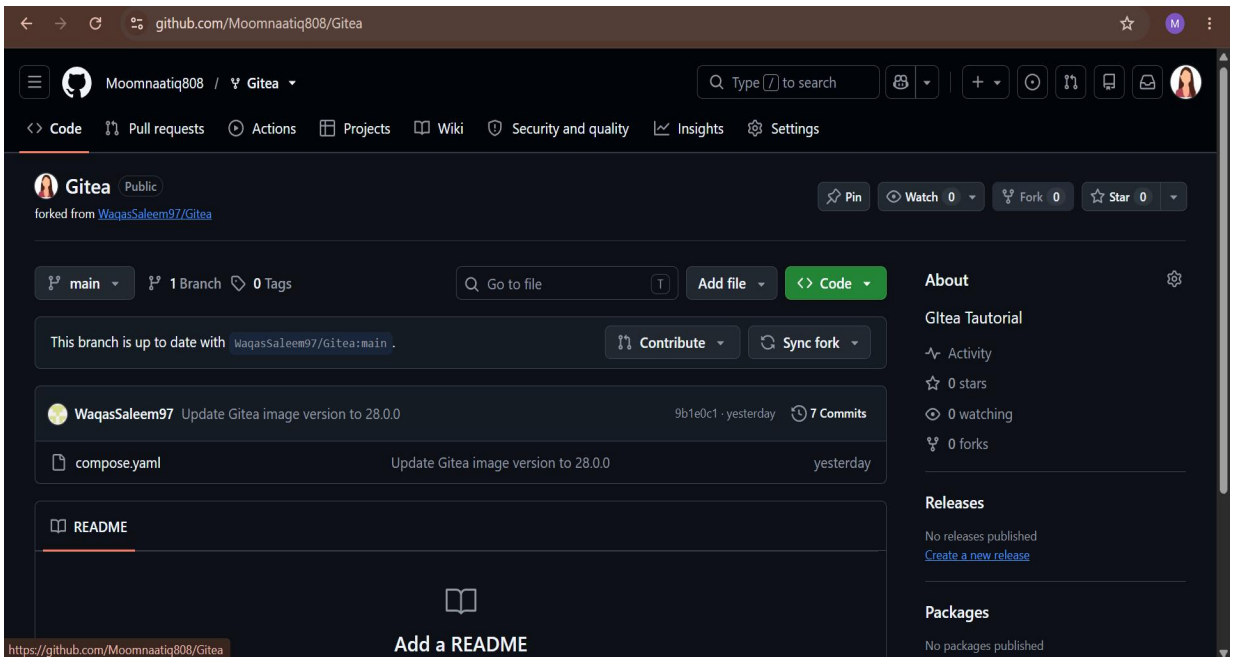
```
Moomnaatiq808@ubuntu:~/DevOps_Lab3_Practice$ git push origin test-force
To github.com:Moomnaatiq808/DevOps_Lab3_Practice.git
 ! [rejected]        test-force -> test-force (non-fast-forward)
error: failed to push some refs to 'github.com:Moomnaatiq808/DevOps_Lab3_Practice.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
Moomnaatiq808@ubuntu:~/DevOps_Lab3_Practice$
```

force_push.png

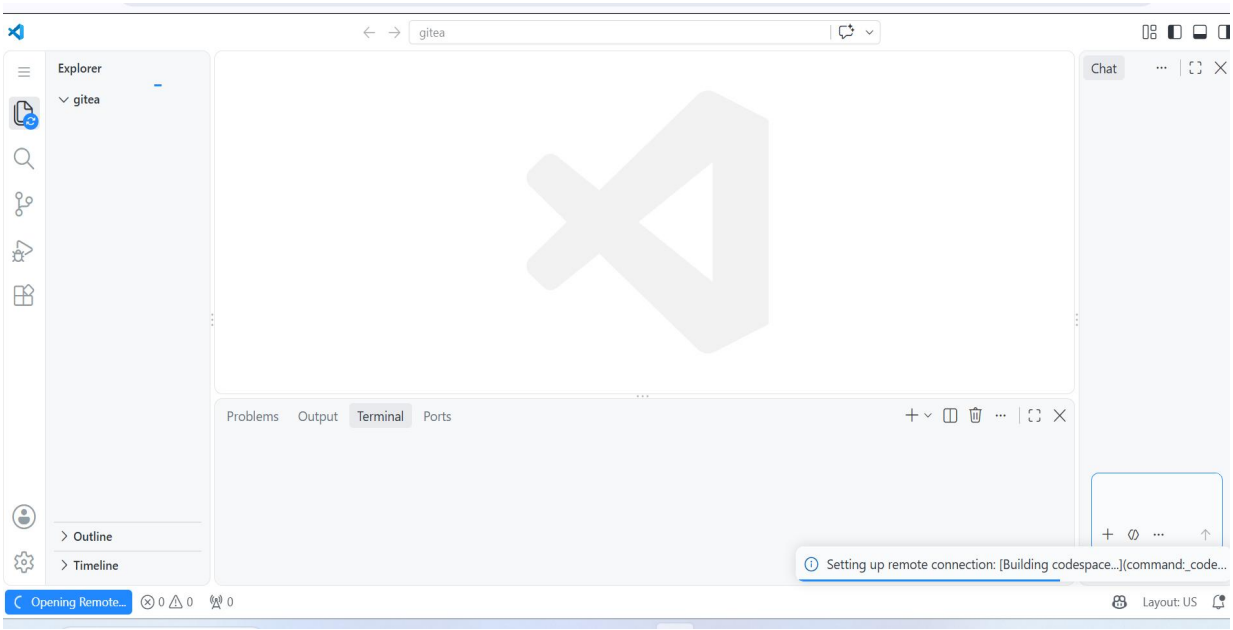
```
Moomnaatiq808@ubuntu:~/DevOps_Lab3_Practice$ git push origin test-force --force
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Moomnaatiq808/DevOps_Lab3_Practice.git
 + 1aa74ff...1bf3247 test-force -> test-force (forced update)
Moomnaatiq808@ubuntu:~/DevOps_Lab3_Practice$
```

Task 10 – Running Gitea in GitHub Codespaces via Docker Compose

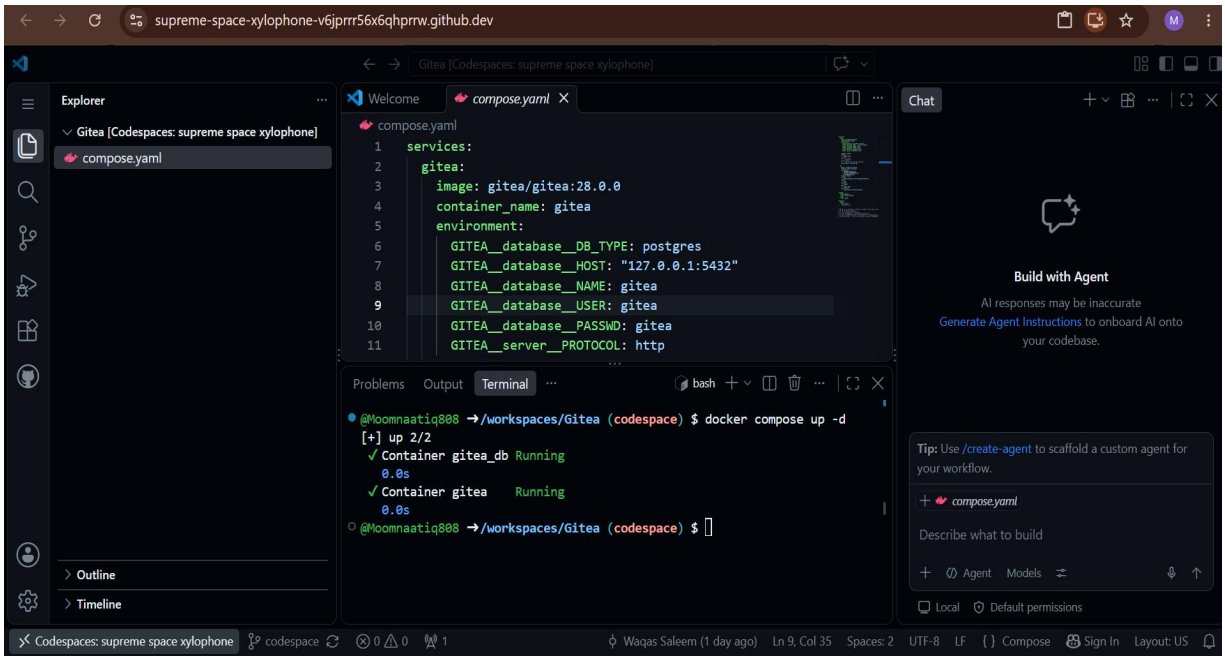
forked_gitea.png



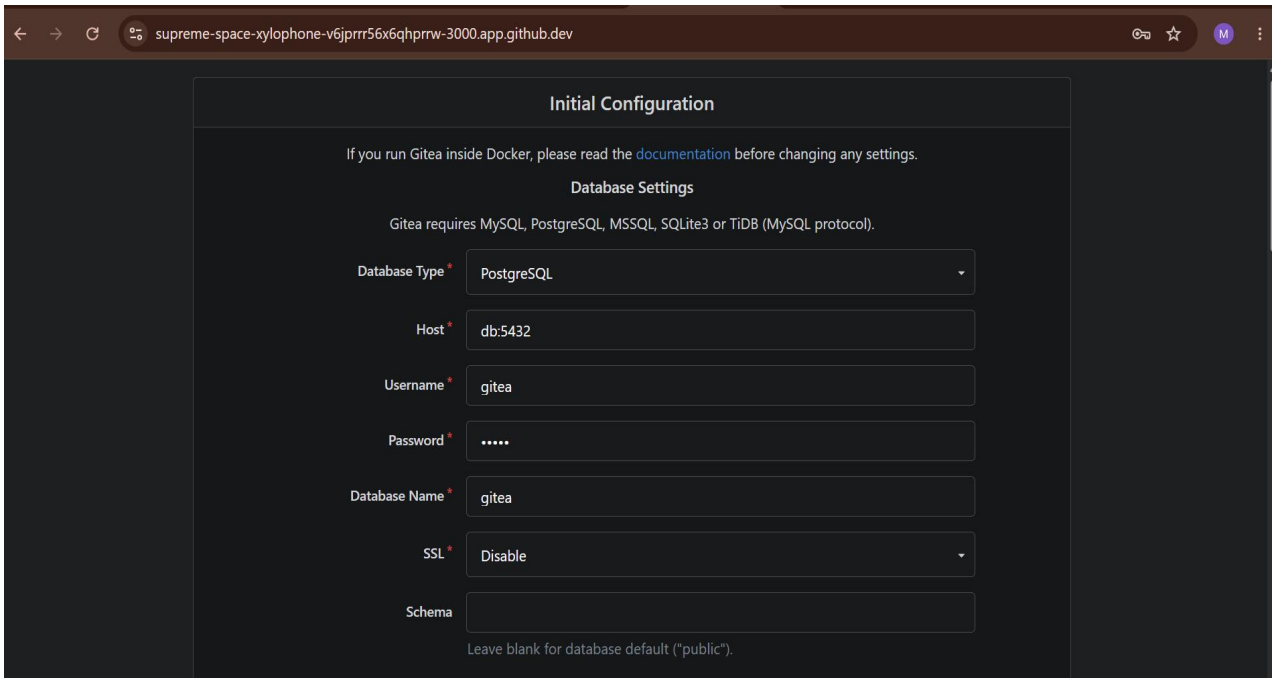
codespace_loading.png



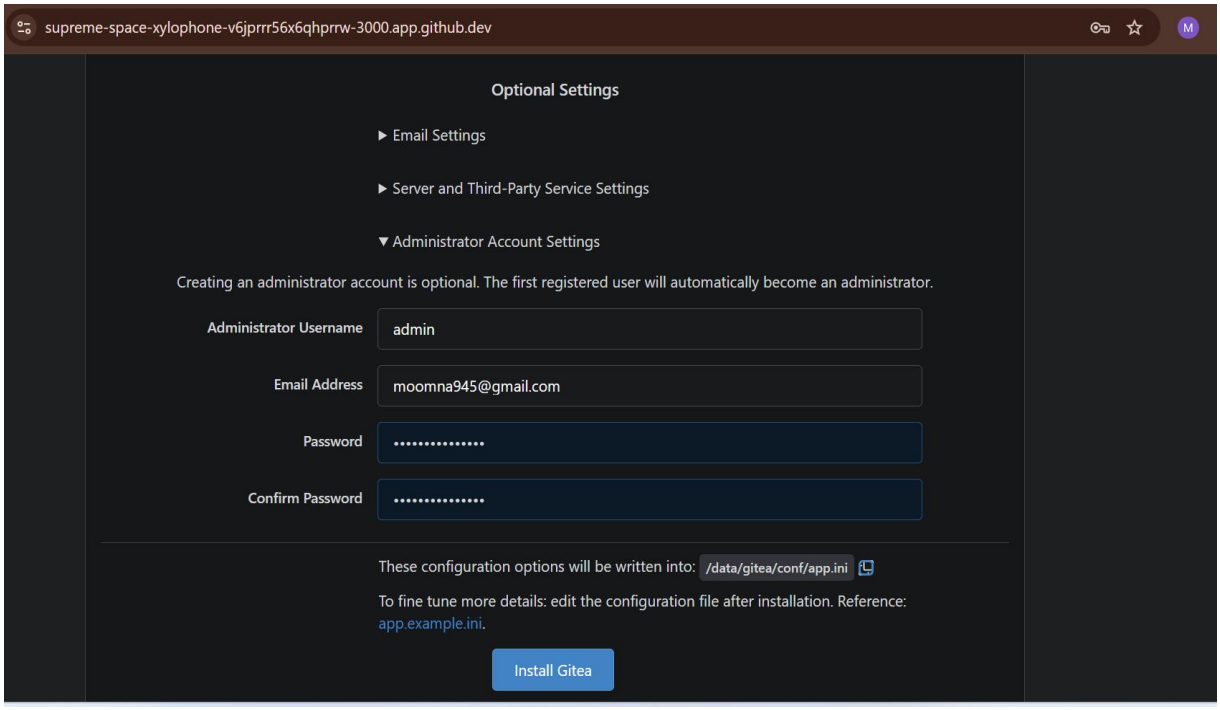
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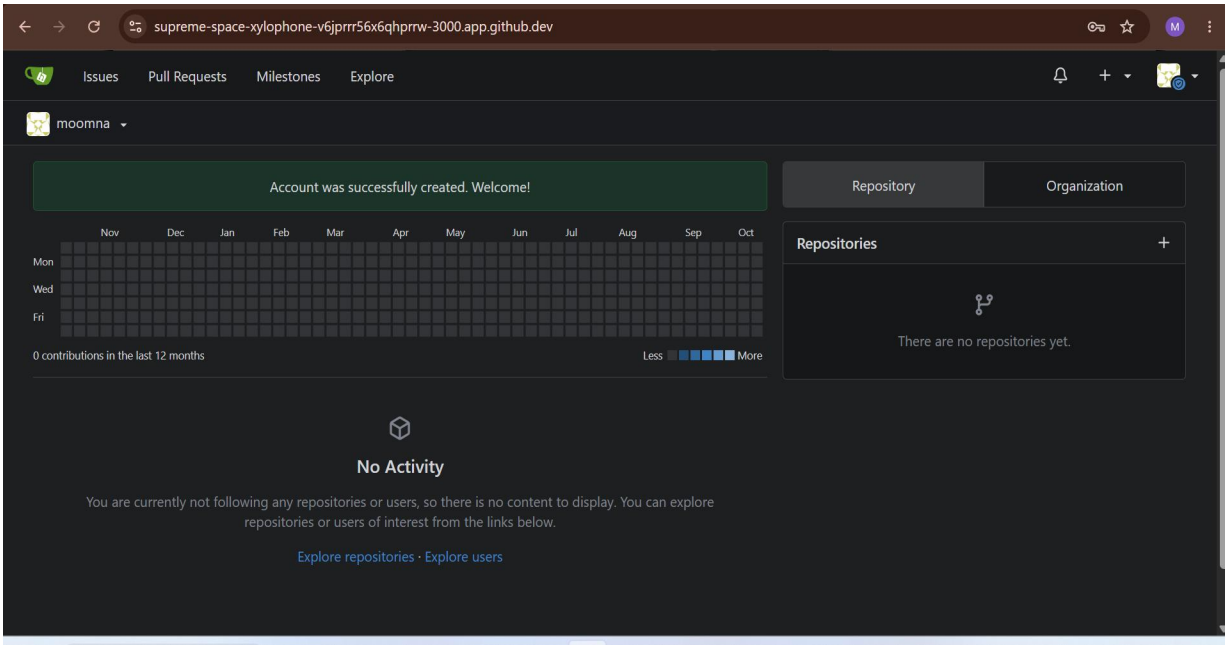
gitea_install_page.png



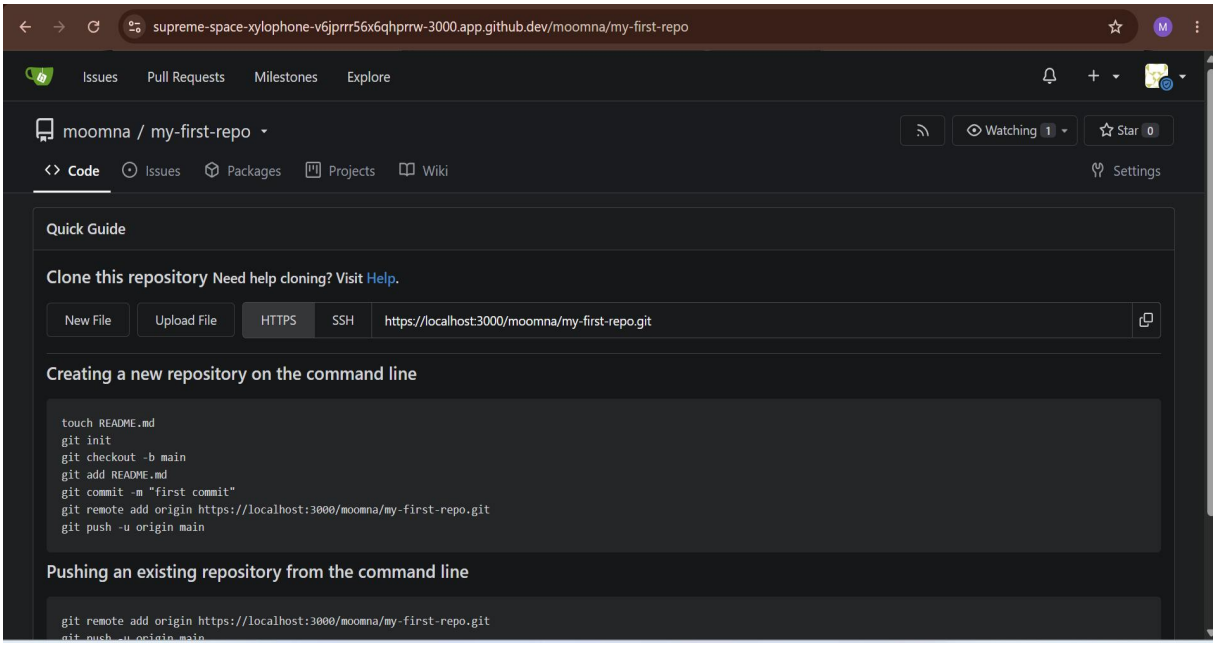
admin_setup.png



gitea_dashboard.png

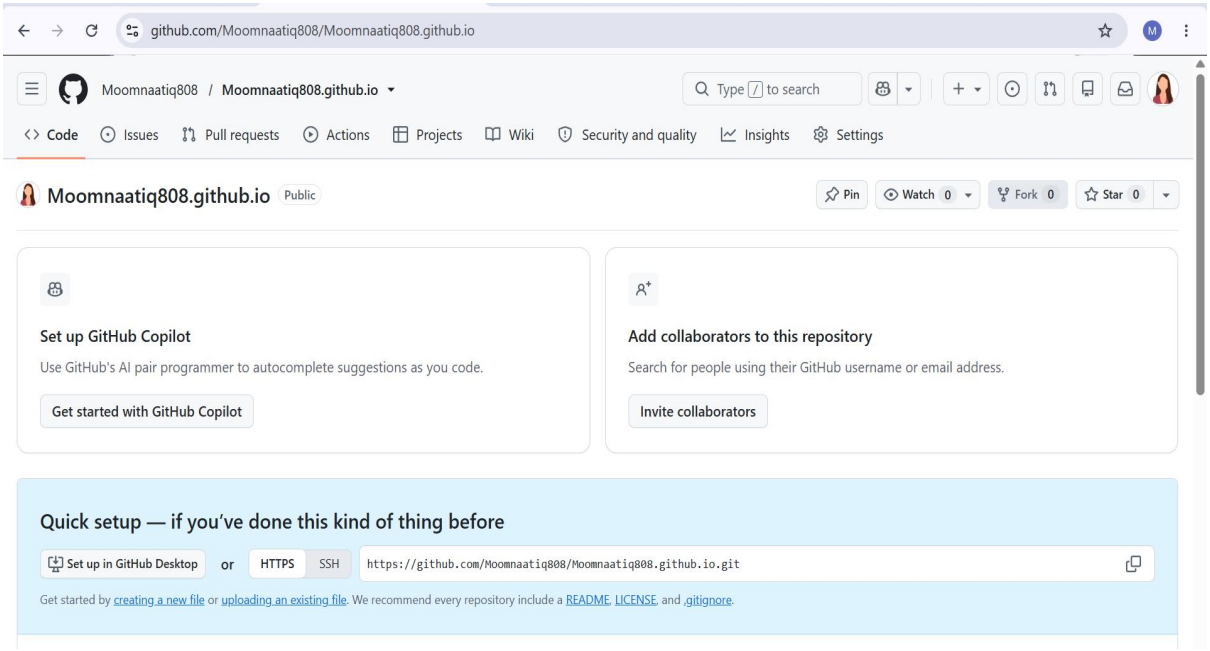


gitea_new_repo.png

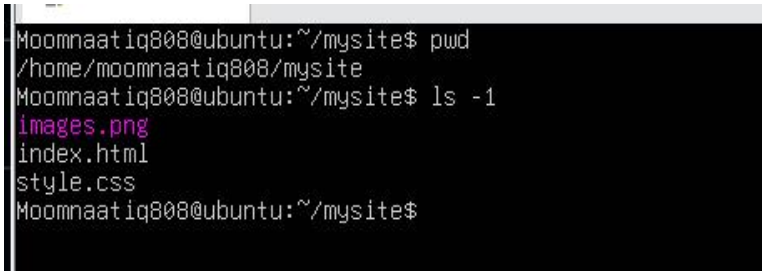


Task 11 – Creating a GitHub Pages Portfolio Site

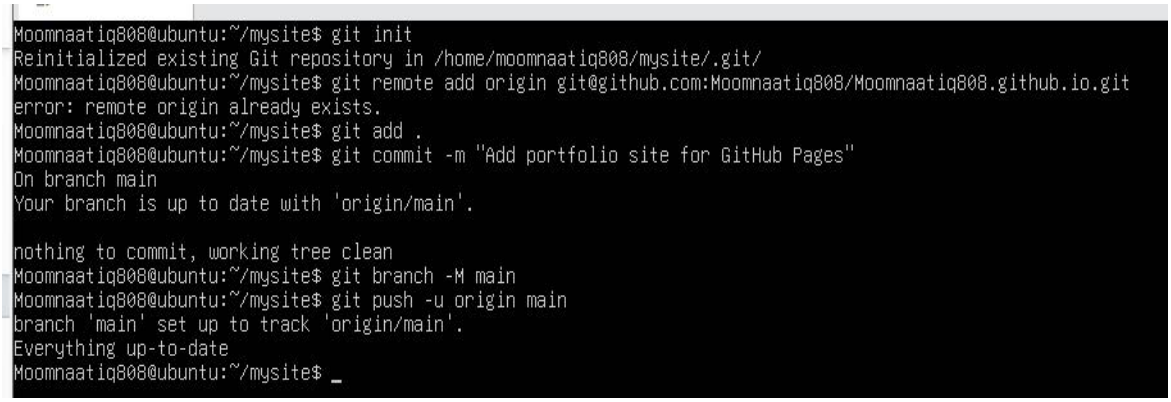
github_pages_repo.png



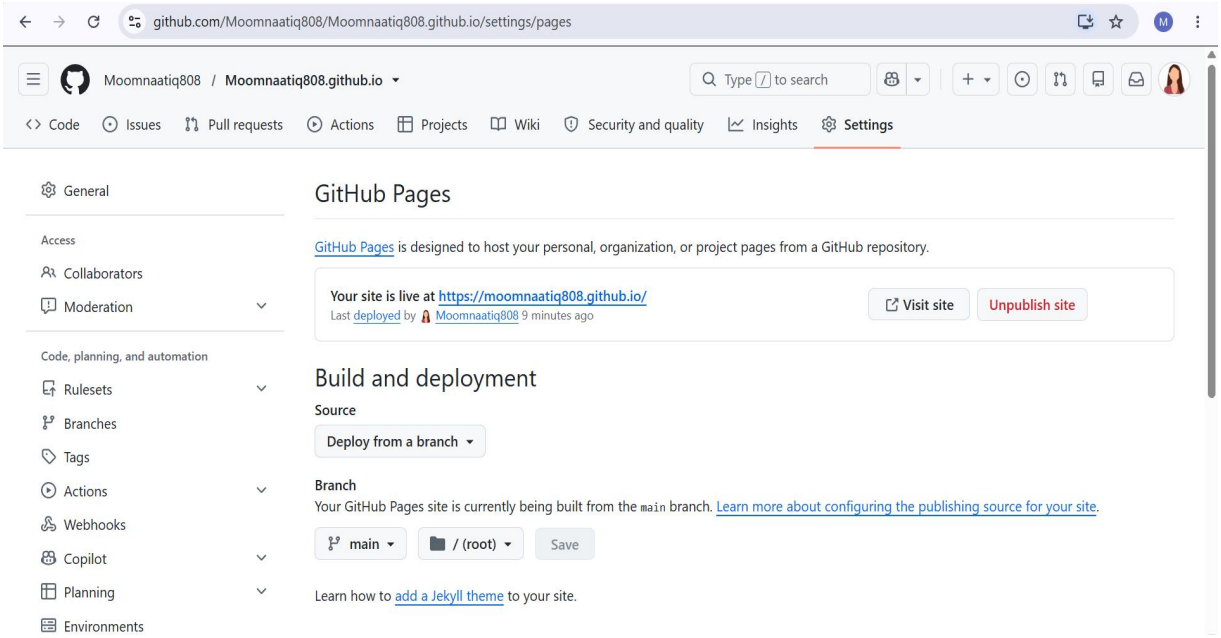
local_static_site.png



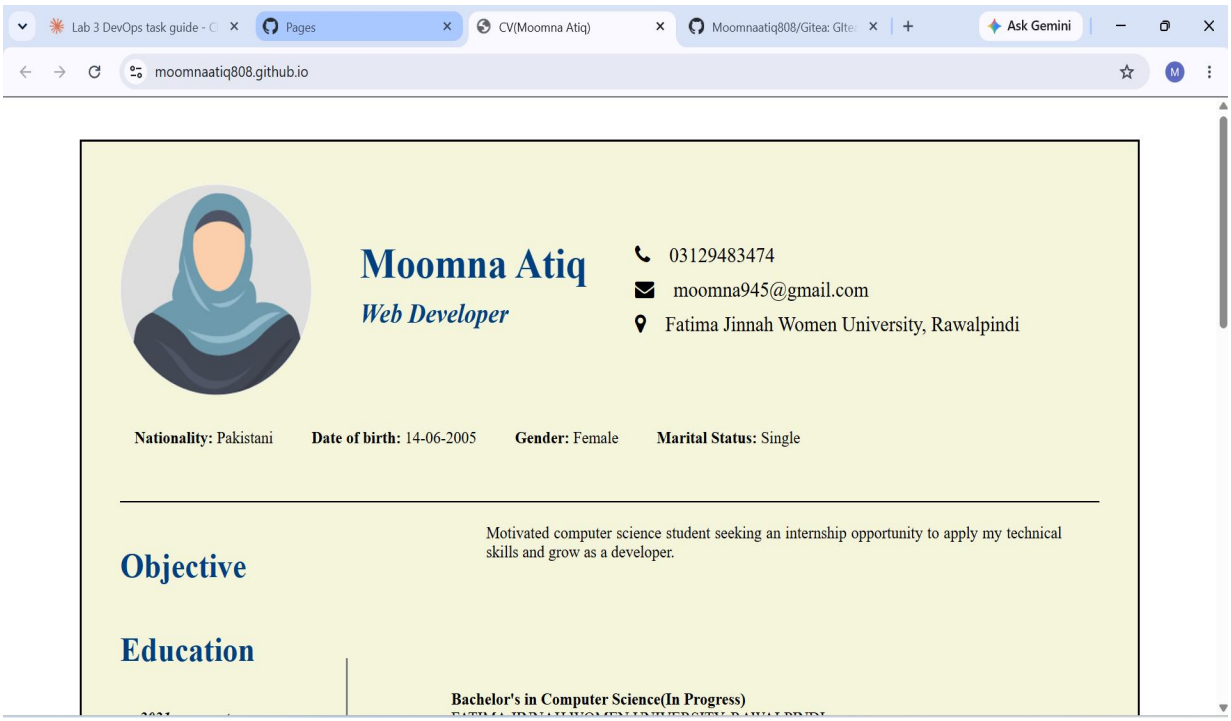
push_static_site.png



github_pages_settings.png



live_site.png


Lab 3 DevOps task guide - C
Pages
CV(Moomna Atiq)
Moomnaatiq808/Gitea: Gite
Ask Gemini
moomnaatiq808.github.io
Moomna Atiq
Web Developer
03129483474
moomna945@gmail.com
Fatima Jinnah Women University, Rawalpindi
Nationality: Pakistani
Date of birth: 14-06-2005
Gender: Female
Marital Status: Single
Objective
Motivated computer science student seeking an internship opportunity to apply my technical skills and grow as a developer.
Education
Bachelor's in Computer Science(In Progress)

End!